Part A: CIFAR-10 Image Generation with GANs and VAEs

Generate 1,000 images across 10 CIFAR-10 classes and determine which generative approach performs best.

Representative CIFAR-10 Training Images by Class

DATASET

50,000 training images
10,000 protected test images
32 × 32 RGB, 10 balanced classes

FINAL OUTPUT

1,000 generated images
100 requested conditions per class

EDA FINDINGS

Strong variation in pose, crop, colour, scale and background
Vehicles had more consistent large-scale structure
Several animal classes were visually more variable and ambiguous

Challenge: at 32 × 32 resolution, realistic colour and texture do not guarantee recognisable class structure.

WORKFLOW

EDA → GAN development → VAE development → freeze both models → protected final comparison → final 1,000 images

Two Generative Approaches, One Fair Evaluation Protocol

Both model families were developed independently, frozen, then compared under the same final protocol.

GAN TRACK

100-D latent z
+ class embedding
→ concatenate
→ Generator
→ Generated image

Discriminator uses image + class map
for adversarial feedback

VAE TRACK

Image + one-hot class
→ Encoder
→ 64-D latent z

64-D z + one-hot class
→ Decoder
→ Reconstruction / new sample

FID ↓

Distributional similarity

BLIND QUALITY

Clear / Marginal / Nonsense

CLASS ADHERENCE

Correct / Ambiguous / Incorrect

DIVERSITY & ORIGINALITY

Duplicates + nearest-neighbour audit

FINAL FID PROTOCOL

10,000 real test images · 5,000 generated images/model/repeat · 3 repeats

50k TRAIN → DEVELOP → FREEZE BOTH MODELS + PROTOCOL → OPEN 10k TEST → FINAL COMPARISON

SELECTION RULE

No single metric selected the final model. We used distributional, visual, class-control, diversity and reproducibility evidence together.

Systematic GAN Development: Why G10 Was Retained

Each experiment addressed a specific weakness, and unsupported changes were rejected.

Experiment decision trail

Question tested	Decision
Dense GAN / DCGAN / WGAN-GP	DCGAN family
Latent dimension	100
Upsampling & capacity	Selected DCGAN retained
Learning-rate recipe	G: 2e-4 / D: 2e-4 retained
Augmentation	Flip + ±2 px retained
Unconditional vs conditional	Conditional G10 retained

FINAL GAN

G10 Conditional DCGAN
Epoch 100

DEVELOPMENT FID

34.8933 ± 0.3457

CLASS AGREEMENT

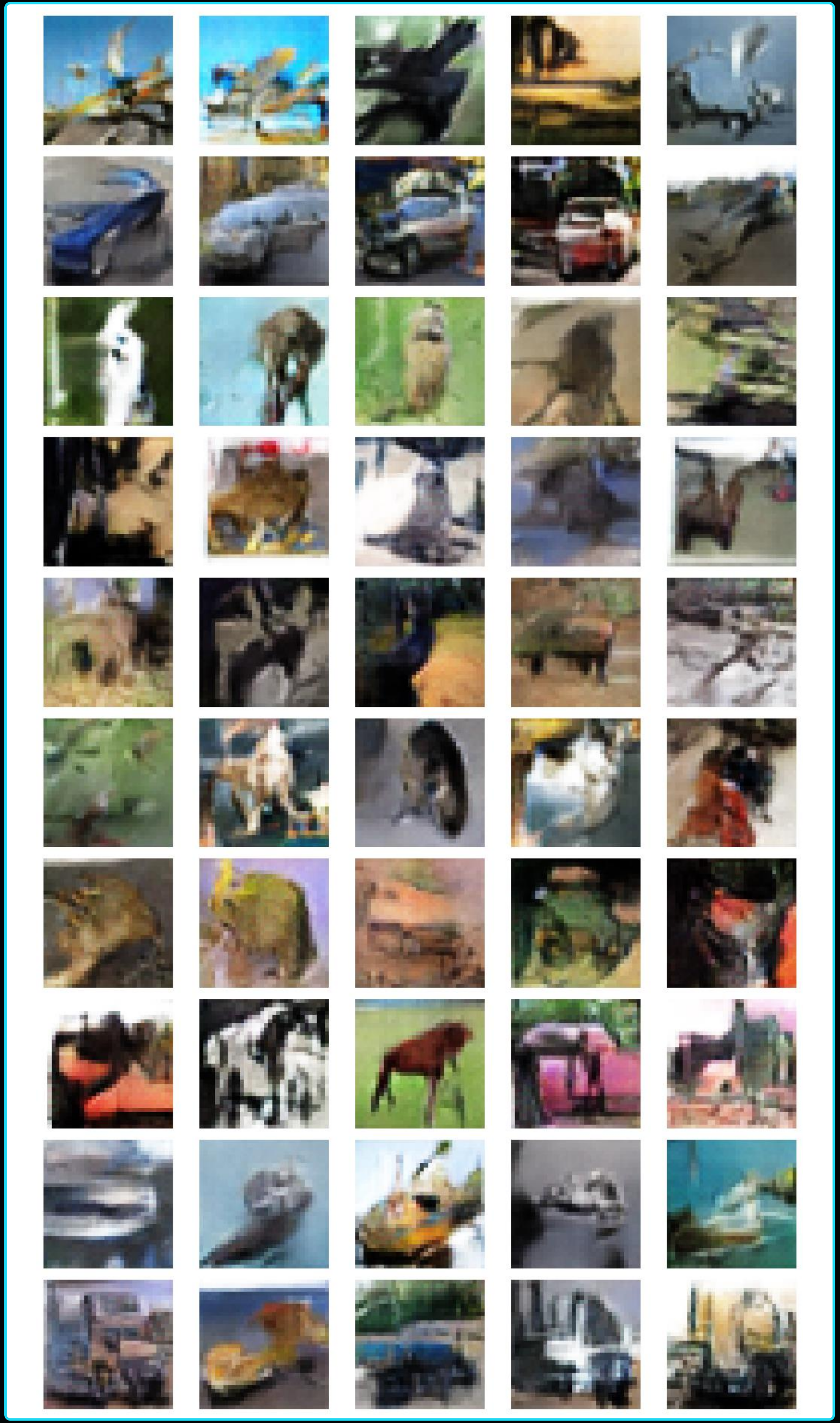
68.9%

MASKED QUALITY

55 Clear / 42 Marginal / 3 Nonsense

SEED 123 CHECK

FID 35.2332 ± 0.1290



G08 and G10 were practically tied on overall quality; G10 was retained because conditioning added meaningful requested-class control.

Systematic VAE Development: Why V04 Was Retained

The final VAE was selected from controlled improvements, not from total loss alone.

Experiment decision trail

Question tested	Decision
Dense / ConvTranspose / UpSampling	**ConvTranspose**
β = 1.0 vs 0.5	**β = 0.5**
Latent 64 vs 128	**64 retained**
Add class conditioning	**Retained**
BCE vs L1 + SSIM	**L1 + SSIM retained**
Adam vs AdamW	**Adam / V04 retained**

FINAL VAE

V04 Conditional ConvTranspose VAE · Epoch 50

DEVELOPMENT FID

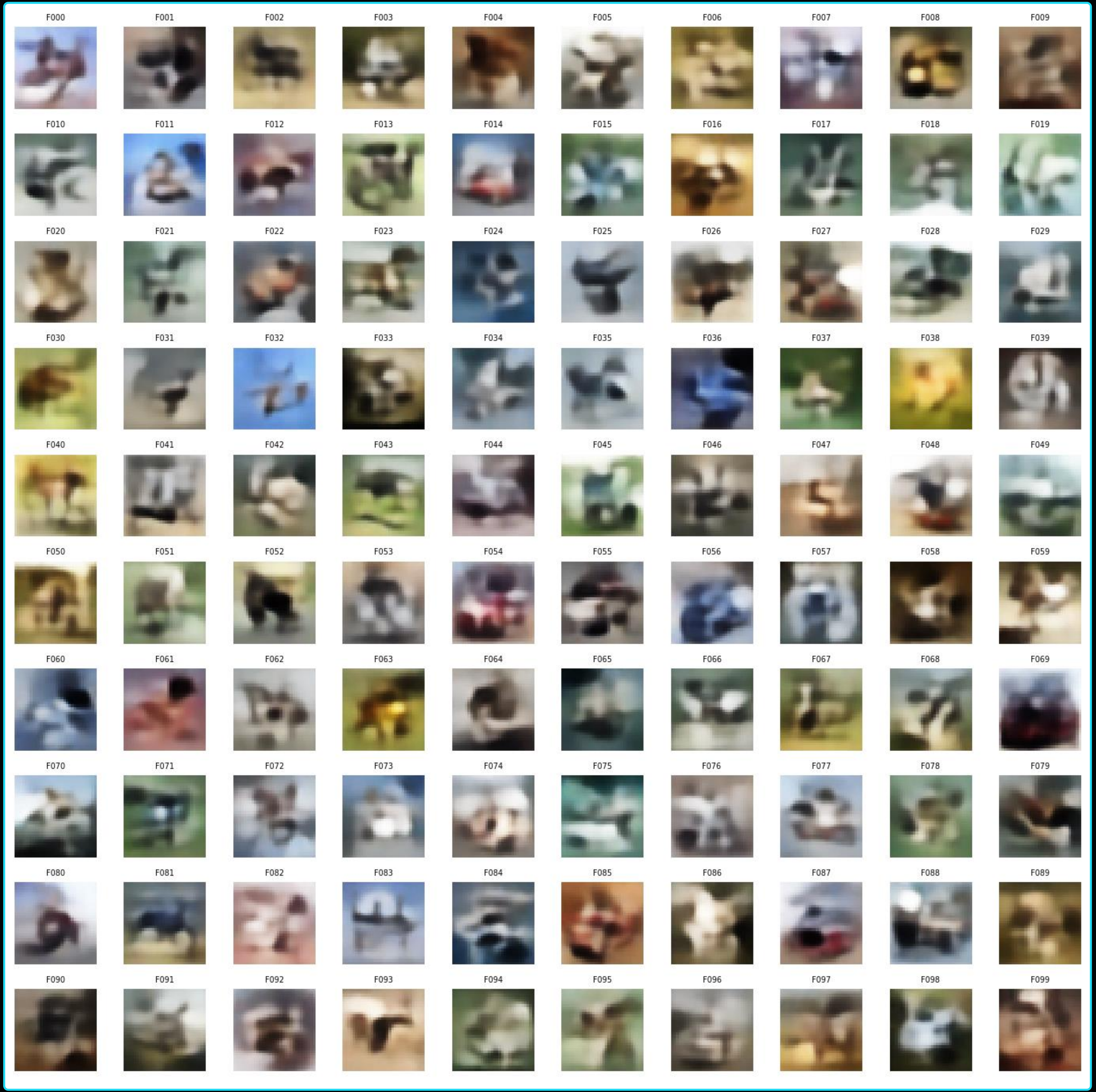
145.5441 ± 0.3302

BLIND QUALITY

0 Clear / 71 Marginal / 29 Nonsense

CLASS AGREEMENT

12%



V05 CAVEAT

AdamW gave only ~0.29% lower FID but needed ~48% more training time. V04 remained the stronger overall trade-off.

Frozen Final Comparison: GAN Clearly Outperformed VAE

Same protected-test reference, same sampling policy, same evaluation definitions.

Final evidence	Conditional GAN	Conditional VAE
Protected-test FID ↓	36.7096 ± 0.1153	147.1899 ± 0.6069
Clear / Marginal / Nonsense	50 / 45 / 5	0 / 71 / 29
Correct / Ambiguous / Incorrect	48 / 30 / 22	1 / 65 / 34
Unique nearest-training neighbours	3,595 / 5,000	1,006 / 5,000
Reload verification	PASS	PASS

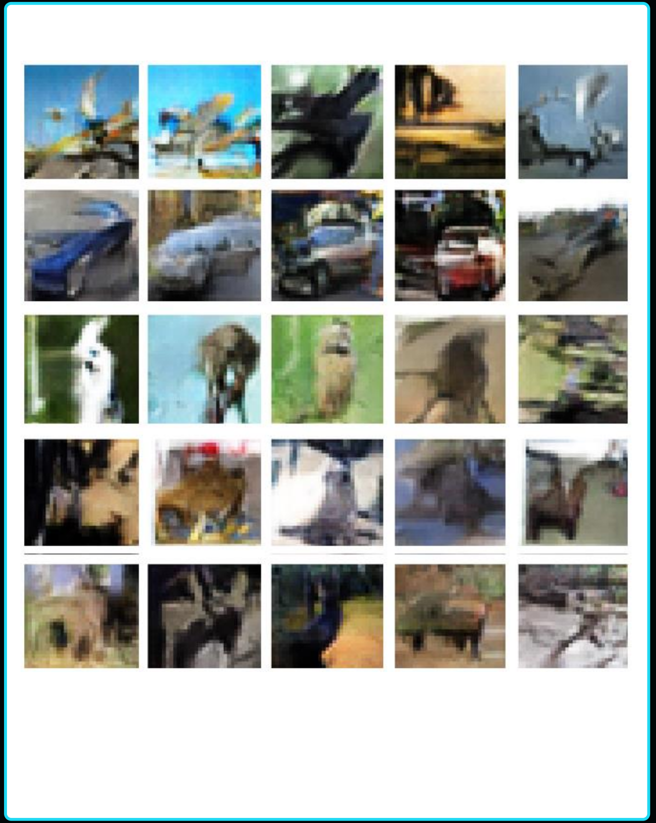
MEMORISATION CHECK

No suspicious training-image proximity
Median distance: test 0.1495 · GAN 0.1856 · VAE 0.1777

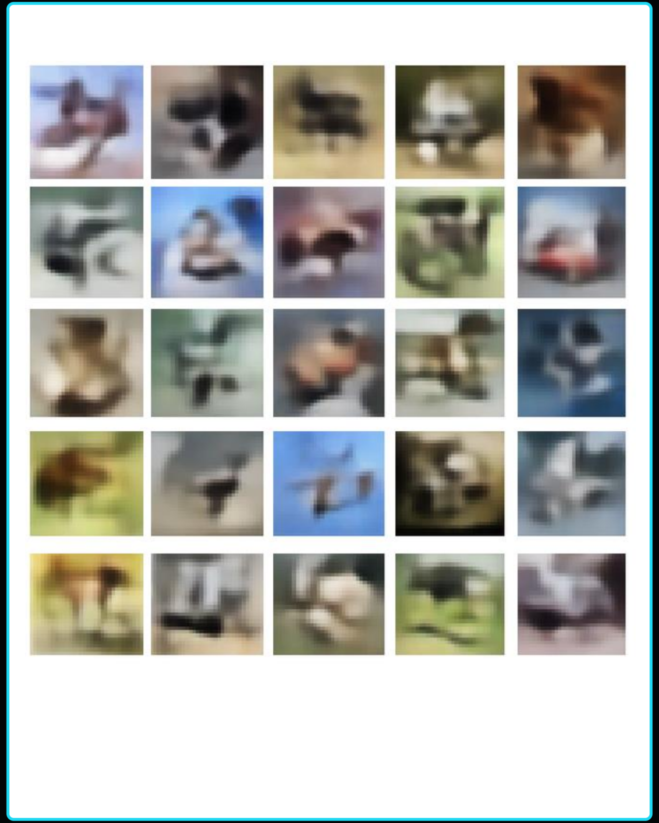
VAE ADVANTAGE

Smaller and faster; useful reconstruction pathway
But weaker for prior-sampled generation

GAN samples



VAE samples



FINAL MODEL: CONDITIONAL GAN

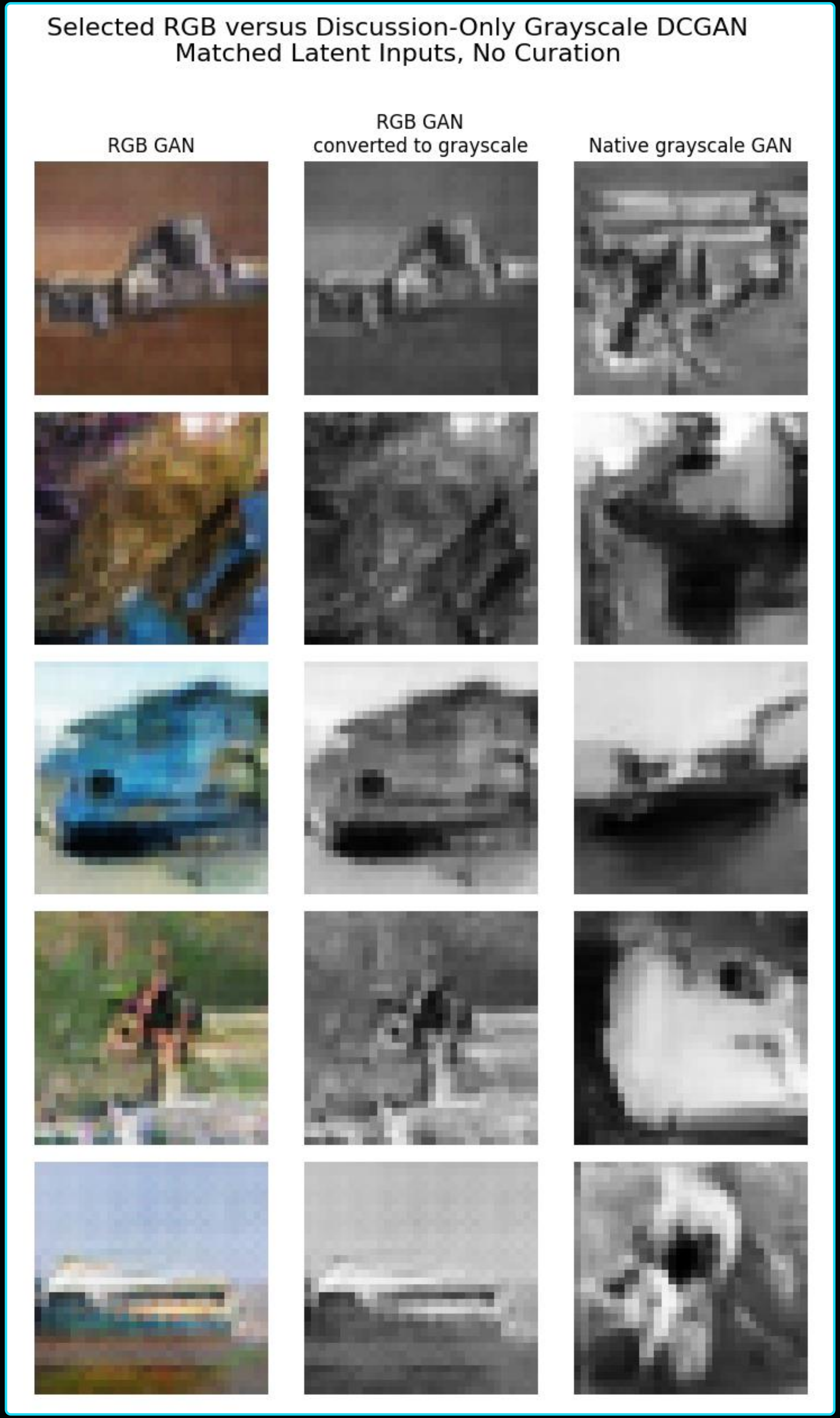
Stronger held-out FID, visual recognisability, class control and generated diversity.

INTERPRETATION

Every primary final evidence dimension pointed in the same direction.

Assignment Questions: What Did the Experiments Show?

The final analysis directly addressed requested-class generation, class difficulty, grayscale and augmentation.



REQUESTED-CLASS GENERATION

GAN: 48 / 100 Correct
VAE: 1 / 100 Correct
Conditioning helps, but labels do not guarantee semantic correctness.

EASIER / HARDER CLASSES

Easier: Automobile 10/10, Ship 8/10, Horse 7/10
Harder: Frog 1/10, Bird 2/10, Dog 2/10, Cat 3/10

RGB VS GRAYSCALE

G12 grayscale FID: 22.0720 ± 0.1403
Manual structural score: 40 / 51 / 9
Grayscale simplified structure but removed useful colour information.

AUGMENTATION

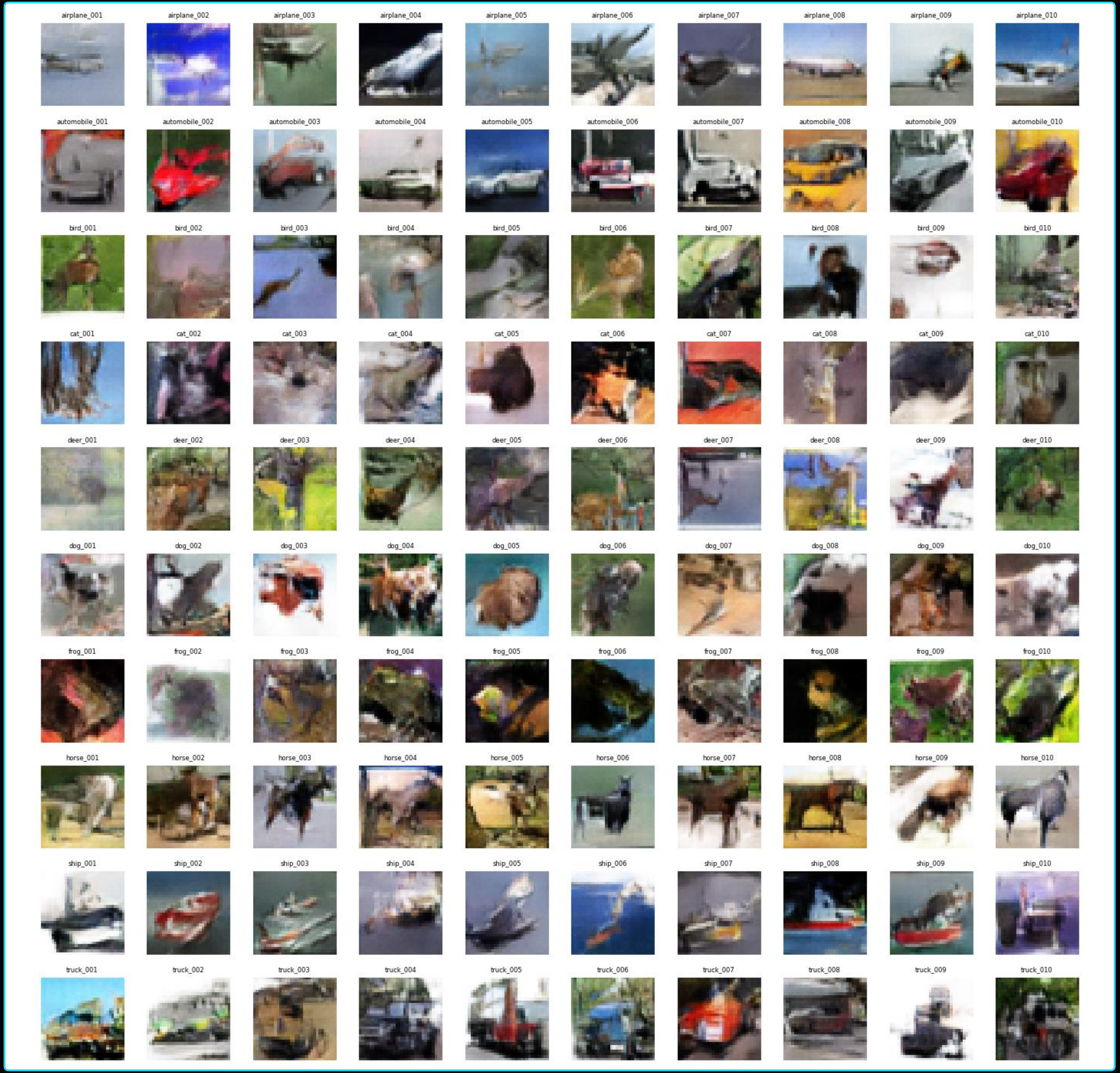
Seed 42: practical tie
Aug 34.8933 vs No-aug 34.6523
Seed 123: no-aug collapsed
Aug 35.2332 vs No-aug 109.2491
Retained for robustness, not a universal claim.

Important: RGB and grayscale FIDs are not directly comparable because they use different real-reference distributions.

The explicit assignment questions were answered through controlled experiments, not after-the-fact claims.

Final 1,000 Images, Reproducibility and Takeaway

The final output was generated from the frozen conditional GAN with no post-generation quality filtering.



FINAL DELIVERABLE

1,000 PNGs
100 requested conditions × 10 classes
RGB · 32 × 32

FILE AUDIT

1,000 readable files
1,000 unique filenames
0 exact duplicate groups

REPRODUCIBILITY

Frozen G10 generator · fresh weight reload verified · final manifest saved

LIMITATION

Filename labels record the requested condition, not guaranteed semantic correctness.

No post-generation quality filtering or cherry-picking.

MAIN TAKEAWAY

The conditional GAN provided the strongest overall balance of distributional similarity, visual quality, class control, diversity and reproducibility.